

MACHINE LEARNING FUNDAMENTALS SERIES

Ridge vs. Lasso Regression: A Visualized Comparison

Understanding L1 vs L2 Penalties Through Parabola Shifts and Kink Discontinuities

Source: StatQuest with Josh Starmer — *Ridge vs Lasso Regression, Visualized!!!*

1. Introduction and Problem Formulation

When fitting linear models to data (such as predicting mouse height from mouse weight), standard Ordinary Least Squares (OLS) minimizes the **Sum of Squared Residuals (SSR)**:

$$\text{SSR}(\beta_1) = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2 \quad (1)$$

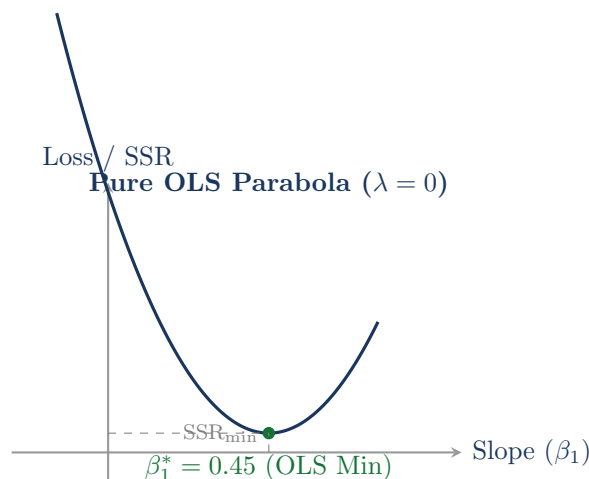
When plotted against various candidate values for the slope β_1 , the SSR curve forms a **smooth, symmetric parabola**. The bottom (vertex) of this parabola represents the OLS optimal slope $\beta_{1,\text{OLS}}^*$ (e.g., $\beta_1^* = 0.45$).

When training data is small or features are many, standard OLS overfits, resulting in high variance. Regularization introduces a penalty term controlled by a hyperparameter $\lambda \geq 0$ to shrink β_1 towards zero. This document visualizes how **Ridge (L_2)** and **Lasso (L_1)** penalties deform the loss parabola differently.

2. Ordinary Least Squares (OLS) Baseline Curve ($\lambda = 0$)

At $\lambda = 0$, both Ridge and Lasso reduce to standard OLS regression:

- Loss Curve:** Pure SSR parabola with a unique global minimum at $\beta_1^* = 0.45$.
- Derivative at Minimum:** $\left. \frac{d(\text{SSR})}{d\beta_1} \right|_{\beta_1^*=0.45} = 0$.
- Overfitting Risk:** On small datasets, $\beta_{1,\text{OLS}}^*$ is inflated due to noise, making test predictions hypersensitive to input fluctuations.



3. Ridge Regression (L_2 Regularization): Visualizing Parabola Shifts

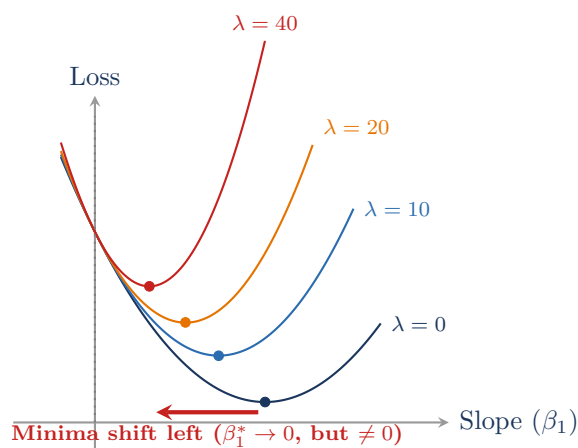
3.1 Mathematical Loss Function

Ridge Regression adds a quadratic penalty proportional to β_1^2 :

$$\text{Loss}_{\text{Ridge}}(\beta_1) = \text{SSR}(\beta_1) + \lambda\beta_1^2 \quad (2)$$

3.2 Visual Behavior as λ Increases

- **$\lambda = 0$** : The loss curve is the original unpenalized OLS parabola centered at $\beta_1^* = 0.45$.
- **$\lambda = 10$** : Adding $10 \cdot \beta_1^2$ adds a symmetric upward parabola centered at 0. The combined curve remains a **smooth parabola**, but its minimum tilts and shifts left to $\beta_1^* = 0.28$.
- **$\lambda = 20, 40$** : As λ grows larger, the penalty term dominates. The minimum continues shifting closer to 0 ($\beta_1^* \rightarrow 0$).
- **Why Ridge Never Reaches Exactly 0**: Because β_1^2 is smooth and differentiable everywhere with derivative $2\lambda\beta_1$, as $\beta_1 \rightarrow 0$, the penalty force vanishes ($2\lambda\beta_1 \rightarrow 0$). Thus, the optimal slope shrinks towards zero asymptotically but **never equals exactly 0** for any finite λ .



4. Lasso Regression (L_1 Regularization): Visualizing the Kink Discontinuity

4.1 Mathematical Loss Function

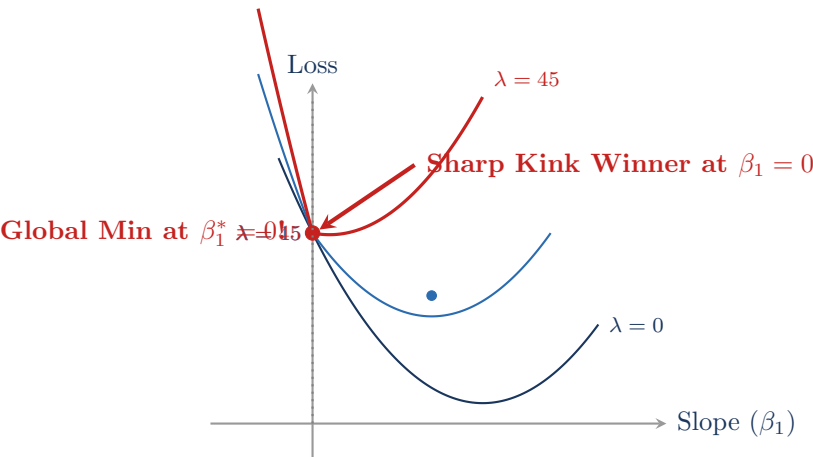
Lasso Regression adds an absolute value penalty proportional to $|\beta_1|$:

$$\text{Loss}_{\text{Lasso}}(\beta_1) = \text{SSR}(\beta_1) + \lambda|\beta_1| \quad (3)$$

4.2 Why Lasso Can Force Slope to Exactly Zero: The "Kink Winner"

- **The Absolute Value Kink**: The function $|\beta_1|$ is non-differentiable at $\beta_1 = 0$, creating a sharp V-shaped point (**kink**).
- **Constant Penalty Force**: The derivative of $\lambda|\beta_1|$ is constant (λ for $\beta_1 > 0$ and $-\lambda$ for $\beta_1 < 0$). Unlike Ridge, the penalizing pull **does not weaken as $\beta_1 \rightarrow 0$** .
- **Kink Dominance for Large λ** : As λ increases, the V-shape at $\beta_1 = 0$ grows steeper. At a critical λ threshold, the sharp kink at $\beta_1 = 0$ becomes the **global minimum of the entire loss curve**.

- ****Automatic Feature Selection****: By setting $\beta_1^* = 0$, Lasso completely removes non-essential predictors from the regression model.



5. Summary Comparison: Ridge vs. Lasso Visualized

Table 1: Side-by-Side Visual and Mathematical Comparison

Feature / Aspect	Ridge Regression (L_2)	Lasso Regression (L_1)
Penalty Term	$\lambda\beta_1^2$ (Quadratic)	$\lambda \beta_1 $ (Absolute Value)
Loss Curve Geometry	Always smooth parabola	Paraboloid with sharp kink at 0
Penalty Derivative near 0	$2\lambda\beta_1 \rightarrow 0$ (Force weakens)	$\pm\lambda$ (Constant force maintained)
Behavior as $\lambda \rightarrow \infty$	Shrinks slope $\rightarrow 0$, but $\neq 0$	Forces slope exactly to 0
Feature Selection	No (Retains all variables)	Yes (Removes useless variables)
Model Type	Dense Model	Sparse Model
Best Use Case	Most features are relevant	Many irrelevant / noisy features

Core Takeaway

While both Ridge and Lasso reduce model variance by shrinking parameters, **Ridge keeps all variables in the equation** by smoothly shrinking slopes near zero. In contrast, **Lasso creates a sharp kink at zero** that acts as a threshold, completely eliminating irrelevant features and producing simpler, interpretable sparse models.